

Robust SE's for glmer

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1 Introduction

The goal is to write a function which will provide robust variances (and possibly small sample size adjustments) for generalized linear mixed models fit using the glmer function in R.

```
vcovCR.glmerMod(obj, cluster, type="classic")
```

obj Fitted glmer model for which to calculate the robust variance-covariance matrix

cluster Optional expression or vector indicating which observations belong to the same cluster. If not specified, will be set to `attr(obj, "cluster")`.

type Character string specifying which robust variance form should be used, with available options "classic" (default), "DF", "KC", "MD", "FG" or "MBN". See "Details" for further information.

2 Methods

The following is adapted from the SAS documentation for the GLIMMIX procedure [6].

Write the generalized linear mixed model as

$$E(Y|\gamma) = g^{-1}(X\beta + Z\gamma) = g^{-1}(\eta) \quad (1)$$

$$Var(Y|\gamma) = \Sigma \quad (2)$$

where Y is $n \times 1$, X is $n \times p$, Z is $n \times q$, and $\gamma \sim N(0, R(\theta))$. Note that it is important to incorporate the number of observations into Σ when data are entered in binomial (as opposed to Bernoulli) format. Define

$$\Delta = \left(\frac{\partial g^{-1}(\eta)}{\partial \eta} \right)_{\beta, \gamma} \text{ a diagonal matrix} \quad (3)$$

$$P = \Delta^{-1}(Y - g^{-1}(\eta)) + X\beta + Z\gamma \quad (4)$$

$$V = ZR(\theta)Z' + \Delta^{-1}\Sigma\Delta^{-1} \quad (5)$$

$$e_i = P_i - X_i\beta \quad (6)$$

See table 1. Note that $(X'V^{-1}X)^{-1}$ is the usual model-based variance.

The robust variance of $\hat{\beta}$ is given by the expression

$$c \times (X'V^{-1}X)^{-1} \left(\sum_g A_g X_g' V_g^{-1} F_g' e_g e_g' F_g V_g^{-1} X_g A_g \right) (X'V^{-1}X)^{-1} + \delta_m \phi (X'V^{-1}X)^{-1} \quad (7)$$

where the sum is over the m independent groups (clusters) and all terms are evaluated at the estimates of $\hat{\beta}$, $\hat{\theta}$, and $\hat{\Sigma}$. See table 2 for values for c , F , A , δ_m , ϕ . Note that for the classical sandwich estimator, $c=1$ and $F = A = I$.

Table 1: Values of Δ for various choices of g

g	Δ
identity	1
logit	$\frac{\exp(\eta)}{(1+\exp(\eta))^2}$
log	$\exp(\eta)$

Table 2: Choices of c, A, F, δ_m , ϕ for various robust variances. $d \geq 1$, $0 \leq r_{fg} < 1$ and $0 \leq r_{mbn} \leq 1$ are user-defined. The user may also chose to set $c = 1$ for the MBN robust variance.

	c	A_g	F_g	δ_m	ϕ
classical	1	I	I	0	0
DF	$\frac{m}{m-p}$	I	I	0	0
Kauermann and Carroll [3]	1	I	$(I - H'_g)^{-1/2}$	0	0
Mancl and DeRouen [4]	1	I	$(I - H'_g)^{-1}$	0	0
Fay and Graubard [1]	1	$\text{diag} \{ (1 - \min(r_{fg}, [Q_g]_{jj}))^{-1/2} \}$	I	0	0
Morel, Bokossa, and Neerchal [5]	$\frac{n-1}{n-p} \times \frac{m}{m-1}$	I	I	$\frac{p}{m-p}$ if $m > (d+1)p$ $\frac{1}{d}$ otherwise	$\max(r_{mbn}, \frac{\text{trace}(\Omega)}{p^*})$

$$H_i = X_i(X'V^{-1}X)^{-1}X'_iV_i^{-1}$$

$$Q_i = X'_iV_i^{-1}X_i(X'V^{-1}X)^{-1}$$

$$\Omega = (X'V^{-1}X)^{-1} \left(\sum_g X'_gV_g^{-1}e_g e'_g V_g^{-1}X_g \right)$$

$p^* = p$ if $m > p$, otherwise p^* equals the number of nonzero singular values of Ω .

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References

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- [6] SAS GLIMMIX documentation (https://documentation.sas.com/doc/en/statug/15.2/statug_g